Car Anti-Theft IoT System

Complete Wiring Diagram and Feature Documentation

1. System Overview

The system is designed to provide robust anti-theft security for vehicles. It uses a microcontroller to manage states, sensors, output devices, and communication interfaces. The system has ARM/DISARM capability, kill-switch logic, GSM mock trigger, OLED for status display, and EEPROM for persistent state storage.

2. Wiring Diagram Overview

The microcontroller (e.g., ESP32 or Arduino) is connected to:

- OLED display (I2C)
- Relay module to simulate kill switch
- Button 1 (ARM/DISARM toggle)
- Button 2 (mock GSM trigger)
- Buzzer for feedback
- TTP223 touch sensor (to be added in hardware phase)
- EEPROM (internal)
- DHT11 sensor (replaced by human/touch detection sensor in real world)
- GSM SIM800L (planned via AT commands)

3. Hardware Wiring Details

- OLED (SDA -> D21, SCL -> D22, VCC -> 3.3V, GND -> GND)
- Relay (IN -> D12, VCC -> 5V, GND -> GND)
- Buzzer (SIG -> D14, VCC -> 5V, GND -> GND)
- Button 1 (one side to D4, other to GND)
- Button 2 (one side to D5, other to GND)
- EEPROM (internal to ESP32)
- Touch Sensor (to be connected to D27 in hardware)

- SIM800L (TX -> RX2, RX -> TX2, powered separately with level shifter)

4. GSM Integration Plan

We plan to use the SIM800L module for GSM functionality. This module communicates via AT commands over Serial2. The system will parse responses from GSM to determine if an ARM/DISARM SMS was received.

5. Future Enhancements

- Real touch detection using TTP223 or capacitive sensor
- Enclosure with tamper detection
- Cloud logging of ARM/DISARM history
- Integration with mobile app (Bluetooth/Wi-Fi)